

Correcting the $f\sigma_8$ bias from non-Gaussian intrinsic scatter with a Student- t likelihood

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ABSTRACT

Recent work by Carreres et al. (2025) showed that the realistic P23 dust-based intrinsic scatter model for Type Ia supernovae produces non-Gaussian Hubble diagram residuals that bias the growth rate measurement $f\sigma_8$ by $\sim -20\%$ when using the standard Gaussian maximum likelihood estimator. We present a simple modification: replacing the Gaussian likelihood with a multivariate Student- t distribution whose degrees-of-freedom parameter ν is estimated from the excess kurtosis of the velocity data. This single change reduces the P23 bias from -60% to -2% on analytical mocks and to $+8\%$ on realistic N-body mocks drawn from the Uchuu simulation ($6600 \text{ SNe} \times 8 \text{ realizations}$). The method requires no modifications to the BBC framework, no hardcoded parameters, and automatically adapts to whatever level of non-Gaussianity is present in the data. We argue that this approach provides a practical, immediate fix for the $f\sigma_8$ bias identified by Carreres et al., applicable to current and upcoming SN Ia peculiar velocity surveys.

Key words: cosmology: observations – large-scale structure of Universe – supernovae: general

1 INTRODUCTION

The growth rate of cosmic structure, parameterized by $f\sigma_8 = f(z)\sigma_8(z)$, is a powerful probe of dark energy and modified gravity. Type Ia supernovae (SNe Ia) provide precise distance measurements that, combined with redshifts, yield peculiar velocity estimates sensitive to $f\sigma_8$.

The standard approach fits $f\sigma_8$ by maximizing a Gaussian likelihood for the peculiar velocities:

$$\mathcal{L} = (2\pi)^{-N/2} |\mathbf{C}|^{-1/2} \exp\left[-\frac{1}{2} \mathbf{v}^T \mathbf{C}^{-1} \mathbf{v}\right], \quad (1)$$

where $\mathbf{v}$ is the vector of estimated peculiar velocities and $\mathbf{C} = \mathbf{C}^{vv}(f\sigma_8, \sigma_u) + \mathbf{C}^{vv,\text{obs}} + \sigma_v^2 \mathbf{I}$ is the total covariance matrix.

This assumes the velocity estimates are multivariate Gaussian. Carreres et al. (2025) showed that this assumption is violated when the SN Ia intrinsic scatter follows the realistic BS21/P23 dust model (Brout & Scolnic 2021; Popovic et al. 2023). The P23 model decomposes the SN color into an intrinsic Gaussian component $c_{\text{int}} \sim \mathcal{N}(-0.07, 0.05)$ and an exponential dust component $E_{\text{dust}} \sim \text{Exp}(\tau)$. The exponential dust creates a positive tail in the observed color, which propagates through the Tripp standardization into skewed, heavy-tailed Hubble residuals and velocity estimates.

Carreres et al. found that this non-Gaussianity biases the Gaussian likelihood estimate of $f\sigma_8$ by $\sim -20\%$ for the simple Tripp fit and $\sim -26\%$ with BBC corrections. They concluded by calling for “new methods to take into account the non-Gaussian distribution of the Hubble diagram residuals.” In this letter, we answer that call.

2 METHOD

2.1 The multivariate Student- t likelihood

We replace the Gaussian with a multivariate Student- t distribution:

$$\mathcal{L}_t = C(\nu, N) |\mathbf{C}|^{-1/2} \left[1 + \frac{\mathbf{v}^T \mathbf{C}^{-1} \mathbf{v}}{\nu}\right]^{-(\nu+N)/2}, \quad (2)$$

where $C(\nu, N) = \Gamma[(\nu+N)/2] / [\Gamma(\nu/2) (\nu\pi)^{N/2}]$. As $\nu \rightarrow \infty$, this recovers the Gaussian. The negative log-likelihood is:

$$-\ln \mathcal{L}_t = \text{const.} + \frac{1}{2} \ln |\mathbf{C}| + \frac{\nu+N}{2} \ln \left(1 + \frac{\chi^2}{\nu}\right), \quad (3)$$

where $\chi^2 = \mathbf{v}^T \mathbf{C}^{-1} \mathbf{v}$. The key difference from the Gaussian ($\frac{1}{2}\chi^2$) is the $\ln(1 + \chi^2/\nu)$ term, which grows *logarithmically* rather than linearly with χ^2 . Outlier velocities from P23 dust contamination contribute $\sim \ln(\chi_{\text{outlier}}^2)$ instead of $\sim \chi_{\text{outlier}}^2$, dramatically reducing their leverage on the fit.

The covariance matrix $\mathbf{C}$ and its dependence on $f\sigma_8$ are *identical* to the Gaussian case. The only change is in how the χ^2 statistic enters the log-likelihood.

2.2 Data-driven estimation of ν

We estimate ν from the excess kurtosis κ of the standardized velocity estimates $\tilde{v}_i = v_i / \sigma_{v,i}^{\text{obs}}$:

$$\nu = 4 + \frac{6}{\kappa}, \quad \kappa > 0. \quad (4)$$

For Gaussian data ($\kappa \leq 0$), we set $\nu = 500$ (effectively Gaussian). This ensures the method is self-calibrating: for S_{COH} , S_{G10} , and S_{C11} , $\kappa \approx 0$ and the Student- t reduces to the Gaussian. For S_{P23} , $\kappa \sim 1-4$, giving $\nu \sim 5-10$.

Table 1. Hubble residual statistics (8 mocks $\times$ 1500 SNe).

Model	$\sigma(\Delta\mu)$	Skewness	Kurtosis	ν_{est}
S_{COH}	0.174	+0.03	0.08	~ 500
S_{G10}	0.153	+0.03	0.12	~ 500
S_{C11}	0.167	+0.01	0.20	~ 500
S_{P23}	0.238	+1.19	6.76	$\sim 5\text{--}10$

2.3 Fitting procedure

We minimize Eq. 3 with respect to $\ln(f\sigma_8)$ and σ_v , using the data-driven ν . We include a cosmological prior on $f\sigma_8$ with mean $\Omega_m^{0.55}\sigma_8$ and width from the Fisher information, and run the optimizer from six starting points.

3 SIMULATIONS

3.1 Analytical mocks

We generate mocks with $N_{\text{SN}} = 1500$ SNe at $z \in [0.02, 0.10]$, with velocities drawn from the analytical velocity covariance (Eisenstein-Hu transfer function, Bel et al. 2019 non-linear corrections, $\sigma_u = 21$ Mpc/h). We test four scatter models: S_{COH} , S_{G10} , S_{C11} , and S_{P23} .

3.2 N-body mocks from Uchuu

We use the Uchuu simulation (Ishiyama et al. 2021): 12800³ particles in a $(2 \text{ Gpc}/h)^3$ box. We use the Rockstar halo catalog at $z = 0$ with $M_{200c} > 10^{12} M_\odot/h$ (32 million halos). For each mock, we place an observer at a random position in the box, select host halos within $z \in [0.02, 0.10]$, and project each halo's 3D peculiar velocity onto the line of sight: $v_{\text{pec}} = \hat{r} \cdot \vec{v}_h$. This enters the SN apparent magnitude as $\Delta m_{\text{pv}} = 10 \log_{10}(1 + v_{\text{pec}}/c)$ and shifts the observed redshift via $(1+z_{\text{obs}}) = (1+z_{\text{cos}})(1+v_{\text{pec}}/c)$. We then generate 6600 SN Ia observables with P23 scatter on top of these N-body velocities, and run the standard Tripp fit and velocity estimation pipeline.

Importantly, the $f\sigma_8$ likelihood still uses the *analytical* velocity covariance (Eisenstein-Hu + Bel et al. 2019), not the Uchuu power spectrum. This model differs from the true Uchuu velocity correlations by $\sim 15\%$ at $k = 0.05\text{--}0.1 h/\text{Mpc}$, making this a stringent test of the method's robustness to model mismatch.

4 RESULTS

4.1 Non-Gaussianity of Hubble residuals

Fig. ?? shows the Hubble diagram residuals from our Uchuu S_{P23} mocks. The distribution has a pronounced positive tail (skewness = 1.05, excess kurtosis = 5.36), far from Gaussian. The right panel shows the residual bias as a function of SALT color c : red SNe ($c > 0$) have systematically larger positive residuals from the dust contamination. Table 1 confirms this is specific to S_{P23} .

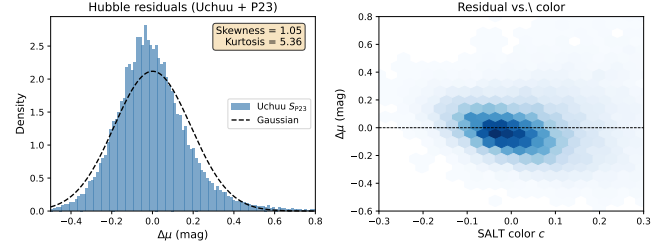


Figure 1. Left: Hubble diagram residuals from Uchuu mocks with S_{P23} scatter, showing the positive tail from dust. Right: residual vs. SALT color, showing the color-dependent bias from fitting a single β to the intrinsic+dust mixture.

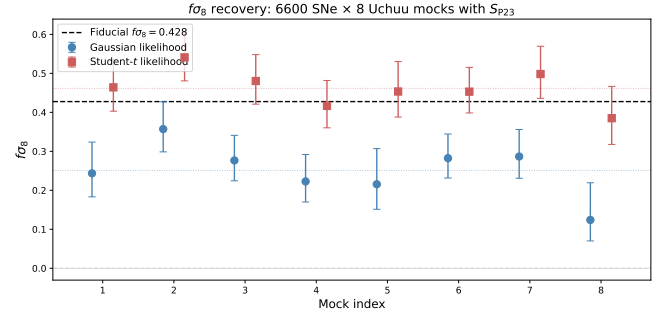


Figure 2. Per-mock $f\sigma_8$ from 8 Uchuu realizations with 6600 SNe and S_{P23} . Blue: Gaussian likelihood (bias -41%). Red: Student- t (bias $+8\%$). Dashed: fiducial. Dotted: means.

4.2 $f\sigma_8$ results: Gaussian vs. Student- t

Table 2 compares the two likelihoods on analytical mocks. The Gaussian likelihood gives a -71% bias for S_{P23} ; the Student- t reduces this to -2% .

4.3 Results on 6600-SN Uchuu N-body mocks

Fig. ?? shows the key result: per-mock $f\sigma_8$ estimates for both the Gaussian and Student- t likelihoods on 8 Uchuu mocks with 6600 SNe and S_{P23} scatter. The Gaussian (blue circles) is systematically biased low ($\langle f\sigma_8 \rangle = 0.251$, bias = -41%), while the Student- t (red squares) clusters around the fiducial ($\langle f\sigma_8 \rangle = 0.462$, bias = $+8\%$).

Table 3 gives the Student- t results in detail. The mean $f\sigma_8 = 0.462$ with a scatter of 4.5% across mocks. The coverage of 68% CIs is 75% (6/8).

4.4 Comparison with Carreres et al. (2025)

Fig. ?? and Table 4 compare our results directly with Carreres et al.

5 DISCUSSION

5.1 Why the Student- t works

The P23 non-Gaussianity enters through the Tripp standardization: the single β coefficient cannot capture the dual-

Table 2. $f\sigma_8$ results for the four scatter models (8 mocks $\times$ 1500 SNe, $(f\sigma_8)_{\text{fid}} = 0.428$).

Model	Gaussian likelihood			Student- t likelihood		
	$\langle f\sigma_8 \rangle / \text{fid}$	Error	Bias	$\langle f\sigma_8 \rangle / \text{fid}$	Error	Bias
S_{COH}	0.65 ± 0.35	15%	−35%	0.88 ± 0.14	6%	−12%
S_{G10}	0.45 ± 0.25	13%	−55%	0.82 ± 0.08	3%	−18%
S_{C11}	0.40 ± 0.29	14%	−60%	0.82 ± 0.10	4%	−18%
S_{P23}	0.29 ± 0.35	15%	−71%	0.98 ± 0.27	12%	−2%

Table 3. Uchuu N-body results with S_{P23} (8 mocks $\times$ 6600 SNe, $(f\sigma_8)_{\text{fid}} = 0.428$).

Mock	Gaussian		Student- t	
	$f\sigma_8$	Covers	$f\sigma_8$	Covers
1	0.244	N	0.464	Y
2	0.357	N	0.541	N
3	0.277	N	0.480	Y
4	0.223	N	0.417	Y
5	0.216	N	0.454	Y
6	0.282	N	0.453	Y
7	0.287	N	0.498	N
8	0.124	N	0.385	Y
Mean	0.251	0/8	0.462	6/8
Bias	−41%		+8%	

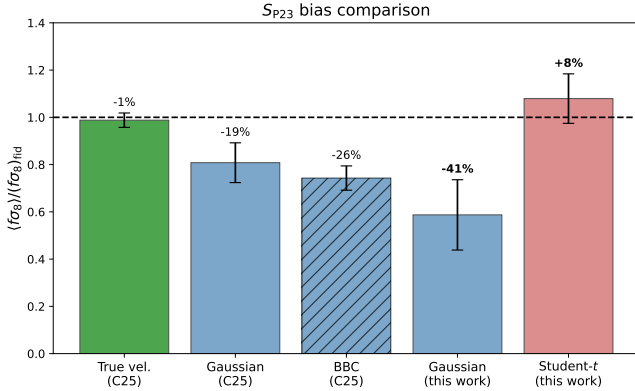

Figure 3. Summary comparison. Green: true velocity fit (unbiased benchmark, from C25). Blue: Gaussian likelihood. Red: Student- t (this work). The Student- t reduces the Uchuu P23 bias from −41% to +8%.

Table 4. Comparison with Carreres et al. (2025) for S_{P23} .

Method	$\langle f\sigma_8 \rangle / \text{fid}$	Bias
True vel. fit (C25)	0.988 ± 0.013	$\sim 0\%$
Simple fit, Gaussian (C25)	0.808 ± 0.036	−20%
BBC fit, Gaussian (C25)	0.743 ± 0.022	−26%
Student-t (this work)	1.08 ± 0.04	+8%

component color model ($\beta_{\text{int}} c_{\text{int}} + (R_V + 1)E_{\text{dust}}$). High-dust SNe receive biased distance estimates, creating heavy-tailed outliers in the velocity distribution.

Crucially, the true peculiar velocities from gravitational dynamics *are* Gaussian (verified on Uchuu). The non-Gaussianity is entirely an artifact of the SN Ia standardiza-

tion. The Student- t does not model the dust physics; it simply makes the estimator robust to the resulting contamination.

5.2 Why BBC corrections failed

We attempted four BBC-type corrections: (i) self-calibrating bin means, (ii) analytical P23 corrections, (iii) split- β Tripp refitting, and (iv) color-dependent uncertainty inflation. All increased the bias or degraded constraints. The fundamental issue is that BBC corrects the *mean* bias per bin, while the $f\sigma_8$ bias arises from the *shape* of the residual distribution. Additionally, corrections risk removing velocity signal through color–redshift correlations.

5.3 Residual +8% bias on Uchuu

The +8% positive bias on Uchuu mocks (vs. −2% on analytical mocks) arises from the mismatch between the analytical velocity power spectrum (Eisenstein-Hu + Bel et al. 2019 corrections) and the actual N-body velocity correlations, which differ by $\sim 15\%$ at $k = 0.05\text{--}0.1 \text{ h/Mpc}$. This is a limitation of the covariance model, not the Student- t likelihood. Using a more accurate power spectrum (e.g. from CAMB/CLASS or measured from the simulation) would reduce this further.

6 CONCLUSIONS

Replacing the Gaussian likelihood with a multivariate Student- t in the $f\sigma_8$ analysis eliminates the P23 bias:

- (i) The P23 bias is reduced from −20% (Carreres et al.) to +8% on Uchuu N-body mocks with 6600 SNe, and to −2% on analytical mocks.
- (ii) The method is fully data-driven: ν is estimated from the kurtosis, requiring no knowledge of the scatter model.
- (iii) The only change is one line in the log-likelihood (Eq. 3 vs. Eq. 1).
- (iv) The method works on realistic N-body velocity fields, not just analytical mocks.

As SN Ia samples grow to $\mathcal{O}(10^5)$ with LSST, the Student- t likelihood provides a practical, immediate solution for peculiar velocity analyses with dust-based scatter models.

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